WORKED SOLUTION

STEP I 2000, Question 10

Mechanics • Collision

STEP I 2000, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q10

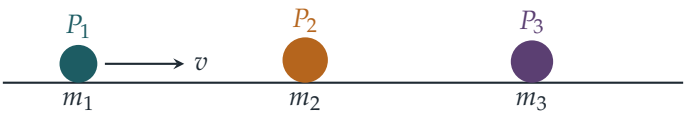
The key idea

Each collision gives us two equations: conservation of momentum and Newton's law of restitution. Here the outcomes are partly known (P_1 stops; P_3 leaves with speed v), so instead of solving for the velocities we solve for the unknown coefficients. The feasibility conditions then come from the one physical constraint on any coefficient of restitution, $0 \leq e \leq 1$, and the final energy becomes a function of the single free mass m_2, which we optimise over the interval those conditions allow.

Solution

Model

We treat the three bodies as particles moving along one line on a smooth table, so no horizontal external forces act and momentum is conserved in each collision. Take the direction of P_1's initial motion as positive.



The first collision

Let P_2 move off with velocity u. Since P_1 is brought to rest,

$$\text{momentum:} \qquad m_1 v = m_2 u \qquad \Longrightarrow \qquad u = \frac{m_1 v}{m_2},$$

$$\text{restitution:} \qquad u - 0 = e(v - 0) \qquad \Longrightarrow \qquad e = \frac{u}{v} = \frac{m_1}{m_2}.$$

The second collision

P_2 now hits the stationary P_3 with speed u. Let P_2's velocity afterwards be w; we are told P_3's is v. Momentum gives

$$m_2u = m_2w + m_3v \quad \Longrightarrow \quad m_1v = m_2w + m_3v \quad \Longrightarrow \quad w = \frac{(m_1 - m_3)v}{m_2},$$

using $m_2u = m_1v$ from the first collision. Restitution (speed of separation $= e' \times$ speed of approach) gives

$$v - w = e'u.$$

Substituting for w and u and multiplying through by m_2/v (which is non-zero):

$$m_2 - (m_1 - m_3) = e'm_1 \quad \Longrightarrow \quad e' = \frac{m_2 + m_3 - m_1}{m_1},$$

which is the required result.

When are such collisions possible?

Any coefficient of restitution lies between 0 and 1. Applying $0 \le e' \le 1$ and multiplying by $m_1 > 0$:

$$0 \le m_2 + m_3 - m_1 \le m_1 \quad \Longleftrightarrow \quad m_1 \le m_2 + m_3 \le 2m_1,$$

which is the inequality asked for. So it is necessary for the collisions to be possible.

? **Ask yourself** Have we used every constraint? What about e?

We also need $0 \le e \le 1$. Since $e = m_1/m_2$ this holds exactly when $m_2 \ge m_1$. That condition is not part of the result we were asked to show (and in fact it makes the left-hand inequality $m_2 + m_3 \ge m_1$ automatic), but it matters in the final part. Putting everything together, the collisions are possible exactly when

$$m_1 \le m_2 \le 2m_1 - m_3.$$

For this interval to be non-empty we need $m_3 \le m_1$. Then $w = (m_1 - m_3)v/m_2 \ge 0$, so after the second collision P_2 keeps moving forwards (or stops) and never comes back to hit P_1, which is at rest. Also $v - w = e'u \ge 0$, so P_2 does not catch P_3 again. These are therefore the only two collisions.

The final energy

Finally P_1 is at rest, P_2 has velocity w and P_3 has velocity v, so the final kinetic energy is

$$E = \tfrac{1}{2}m_2w^2 + \tfrac{1}{2}m_3v^2 = \frac{(m_1 - m_3)^2v^2}{2m_2} + \tfrac{1}{2}m_3v^2.$$

With m_1, m_3 and v fixed, only m_2 can vary, over $m_1 \le m_2 \le 2m_1 - m_3$. The first term is a non-negative constant divided by m_2, so E is a decreasing function of m_2 (constant if $m_1 = m_3$, when the interval is the single point $m_2 = m_1$ anyway).

The largest energy occurs at $m_2 = m_1$ (the first collision is perfectly elastic, $e = 1$):

$$E_{\max} = \frac{(m_1 - m_3)^2 v^2}{2m_1} + \frac{m_3 v^2}{2} = \frac{(m_1^2 - m_1 m_3 + m_3^2)\, v^2}{2m_1}.$$

The smallest energy occurs at $m_2 = 2m_1 - m_3$ (the second collision is perfectly elastic, $e' = 1$):

$$E_{\min} = \frac{(m_1 - m_3)^2 v^2}{2(2m_1 - m_3)} + \frac{m_3 v^2}{2} = \frac{(m_1^2 - 2m_1 m_3 + m_3^2 + 2m_1 m_3 - m_3^2) v^2}{2(2m_1 - m_3)} = \frac{m_1^2 v^2}{2(2m_1 - m_3)}.$$

Checks

- Energy can never increase in collisions, so we need $E_{\max} \leq \frac{1}{2} m_1 v^2$. This is $m_1^2 - m_1 m_3 + m_3^2 \leq m_1^2$, that is $m_3(m_3 - m_1) \leq 0$, true since $m_3 \leq m_1$.
- One can check that $E_{\max} - E_{\min} = \dfrac{(m_1 - m_3)^3 v^2}{2m_1(2m_1 - m_3)} \geq 0$, and both answers equal $\frac{1}{2} m_1 v^2$ when $m_1 = m_3$ (then $m_2 = m_1$, both collisions are elastic and no energy is lost).
- Numerically, with $m_1 = 2$, $m_3 = 1$, $v = 1$: the initial energy is 1, $E_{\max} = \frac{3}{4}$ and $E_{\min} = \frac{2}{3}$. Stepping m_2 through $[2, 3]$ in exact arithmetic reproduces these extremes, and confirms that $e, e' \in [0, 1]$ throughout and fail outside.

Common slips

- Optimising over $m_1 - m_3 \leq m_2 \leq 2m_1 - m_3$ (from e' alone). This gives a "maximum" of $\frac{1}{2} m_1 v^2$, the full initial energy, at a value of m_2 for which $e = m_1 / m_2 > 1$, which is impossible.
- Getting the restitution law the wrong way round: it is the speed of *separation* that equals e times the speed of *approach*.

Beyond STEP

The pattern here, where each constraint carves out an allowed interval for a parameter and we then optimise a monotonic function over that interval, is the simplest case of constrained optimisation, which grows into linear programming and Lagrange multipliers. On the physics side, collisions are much cleaner in the centre-of-mass frame, where the coefficient of restitution simply scales every velocity by $-e$ and the energy lost is $\frac{1}{2}\mu(1 - e^2) \times (\text{relative speed})^2$ with μ the reduced mass. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops this view of collisions.

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